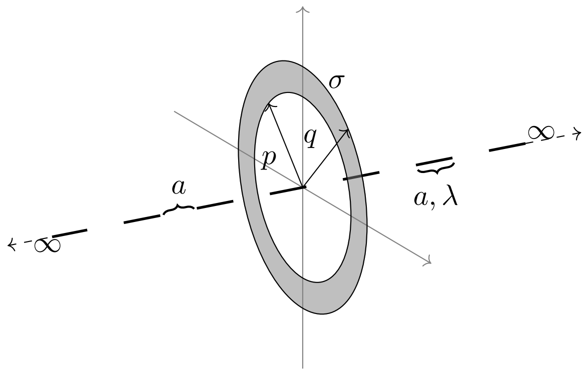
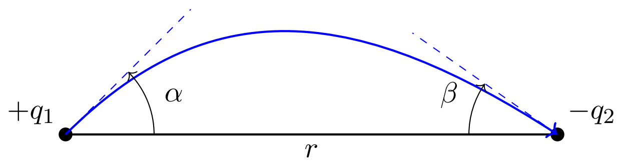
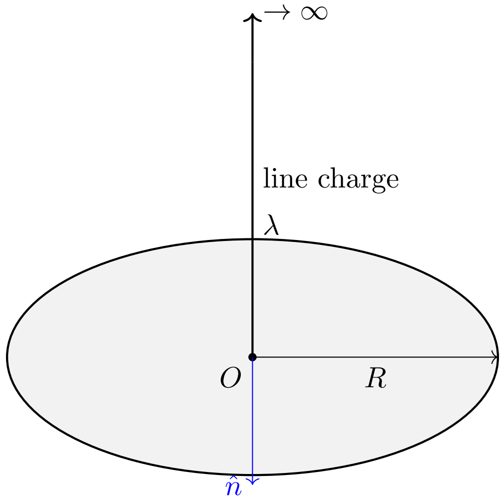
1. Find the net force on ring due to the following line charge distribution. The ring has a uniform charge density σ $\frac{C}{m^2}$, with inner radius p and outer radius q. The length of each line segment is a (in meter), with linear charge distribution λ $\frac{C}{m}$, the distance between each line segment is also a. $\left(\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{\sqrt{a^2+(nb)^2}} = \frac{\pi}{b}\frac{1}{\sinh\left(\frac{\pi a}{b}\right)}\right)$



2. Consider 2 point charges of magnitude q_1 and $-q_2$ placed at a distance r from each other suppose that an electric field line emerges from q_1 at an angle α with the line joining the 2 charges and terminates into $-q_2$ at an angle β. Find the relation governing the 2 angles.



3. Consider a semi-infinite line charge with linear charge density λ C/m which rests on a circular disc of radius R (a circular region of free space). Compute the flux passing through the circular disc.

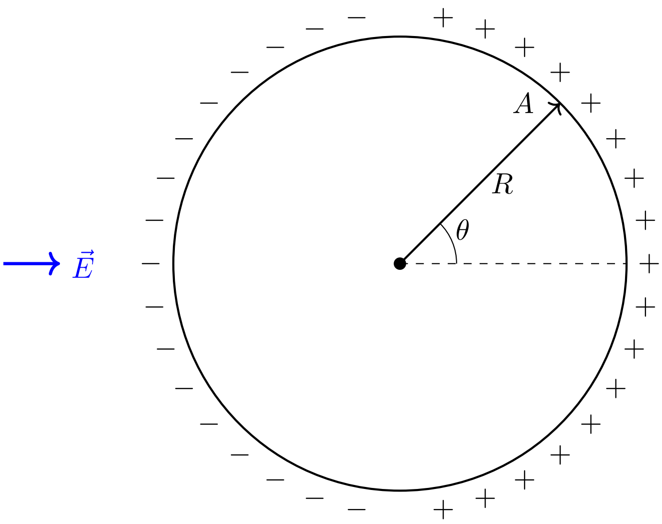


4. Consider the following potential distribution in spherical coordinates

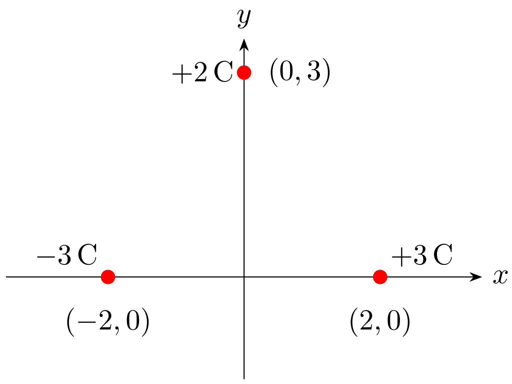
$$\Phi(r,\theta) = \frac{V_0}{a}\left(r - \frac{r^2}{2a} + \frac{a(3\cos^2\theta - 1)}{6}\right)$$

What is the electric field and where is its maximum

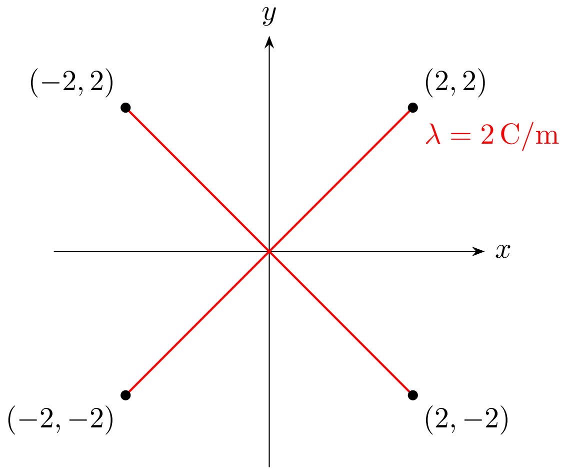
5. A solid conducting sphere of radius R is placed in a uniform electric field $\vec{E}$ as shown in Figure. Due to electric field non-uniform surface charges are induced on the surface of the sphere. Consider a point A on the surface of sphere at a polar angle θ from the direction of electric field as shown in Figure. Find the surface density of induced charges at point A in terms of electric field and polar angle θ.



6. Consider the below charge configurations:



(a) Discrete point charges



(b) X-shaped line charge

(a) For these configurations, evaluate and plot the following quantities at the field points:
   i. Electric field $\vec{E}$
   ii. Electric potential V
   iii. Find out the energy required to assemble the configuration as in configuration (a).

(b) Verify through graphical visualization or representation:
   i Gauss' law
   ii $\vec{E} = -\nabla V$
   iii $\vec{E}$ is conservative

7. A point charge of $12\,\mathrm{nC}$ is located at the origin. Four infinitely long uniform line charges are located parallel to the z-axis in the plane $x = 0$ as follows: Two line charges of density $80\,\mathrm{nC/m}$ at $y = -1m$ and $-\,5\,\mathrm{m}$, and two line charges of density $-50\,\mathrm{nC/m}$ at $y = -2\,\mathrm{m}$ and $-\,4\,\mathrm{m}$.

(a) Determine the total electric flux crossing the plane $y = -3$, and state its direction.

(b) Determine the total electric flux leaving the surface of a sphere of radius $4\,\mathrm{m}$ centered at $C(0, -3, 0)$.